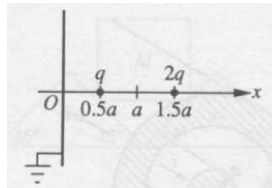
Physics 5020: quiz 5. Student's name:

(Dated: February 9, 2011)

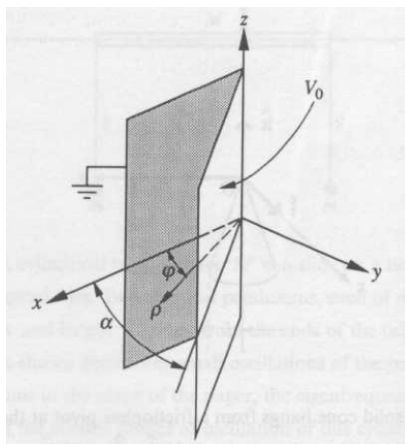
Problem 1 [1 point].

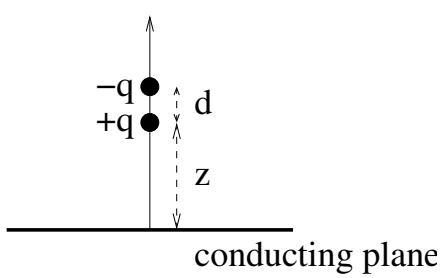
Two positive charges of q and $2q$ coulombs are located on the x-axis at $x = 0.5a$ and $1.5a$, respectively, as shown below. There is an infinite, grounded conducting plane at $x = 0$. What is the magnitude of the net force on charge q?
(A) $\frac{1}{4\pi\epsilon_0}\frac{q^2}{a^2}$ (B) $\frac{1}{4\pi\epsilon_0}\frac{3q^2}{2a^2}$ (C) $\frac{1}{4\pi\epsilon_0}\frac{2q^2}{a^2}$ (D) $\frac{1}{4\pi\epsilon_0}\frac{3q^2}{a^2}$ (E) $\frac{1}{4\pi\epsilon_0}\frac{7q^2}{2a^2}$



Problem 2 [2 points].

Two large conducting plates form a wedge of angle α as shown. The plates are insulated from each other; one has potential V_0 and the other is grounded. Assuming that the plates are large enough so that potential difference between them is independent of the cylindrical coordinates z and ρ, the potential anywhere between the plates as a function of the angle ϕ is
(A) $\frac{V_0}{\alpha}$ (B) $\frac{V_0\phi}{\alpha}$ (C) $\frac{V_0\alpha}{\phi}$ (D) $\frac{V_0\phi^2}{\alpha}$ (E) $\frac{V_0\alpha}{\phi^2}$



Problem 3 [2 points].

An electric dipole of magnitude $p = qd$, made of two point charges $+q$ and $-q$ separated by distance d, is placed above an infinite grounded conducting plane as shown below, and is oriented along the axis normal to the surface. The distance z from the first charge, $+q$, to the plate is much greater than the dipole size, $z \gg d$. The net force experienced by the dipole is
(A) zero (B) $\frac{1}{4\pi\epsilon_0}\frac{3p^2}{(2z)^4}$ upward (C) $\frac{1}{4\pi\epsilon_0}\frac{3p^2}{(2z)^4}$ downward (D) $\frac{1}{4\pi\epsilon_0}\frac{3p^2}{8z^4}$ upward (E) $\frac{1}{4\pi\epsilon_0}\frac{3p^2}{8z^4}$ downward